

LESSON 6-1 • SESSION 1 • PRACTICE • FORM A**Dividing by a Unit Fraction**

Learning target – I can divide a whole number by a unit fraction and explain what the quotient counts. · 6.NS.A.1

REMEMBER

- A **unit fraction** has 1 on top, like $1/3$. Dividing by it asks: **how many pieces of that size fit?**
- **Keep, Change, Flip:** keep the first number, change $\div$ to $\times$, flip the divisor. $4 \div 1/3 = 4 \times 3$.
- The quotient is **greater** than the number you start with. Check: quotient $\times$ divisor = start.

WORKED EXAMPLEA cook has 4 cups of broth. How many $1/3$ -cup servings can she pour?**Step 1** Write the division: $4 \div 1/3$.**Step 2** Keep 4, change $\div$ to $\times$, flip $1/3$ to 3: $4 \times 3 = 12$.**Step 3** Check: $12 \times 1/3 = 4$ cups.**Answer 12 servings****WORDS** unit fraction (*fracción unitaria*) · reciprocal (*recíproco*) · quotient (*cociente*)**PRACTICE***show your work · label units*

- 1.**
- Divide. Think: how many of the unit fraction fit into the whole number?

a. $8 \div 1/2 = \underline{\hspace{2cm}}$

b. $3 \div 1/4 = \underline{\hspace{2cm}}$

c. $10 \div 1/3 = \underline{\hspace{2cm}}$

d. $3 \div 1/5 = \underline{\hspace{2cm}}$

e. $4 \div 1/10 = \underline{\hspace{2cm}}$

f. $7 \div 1/6 = \underline{\hspace{2cm}}$

- 2. Clay and Wax.**
- An art teacher has 8 pounds of clay and a 2-pound block of wax. Each bar is 1 pound.



- a.** Each project uses $1/4$ pound of clay. How many projects? projects
- b.** She cuts the wax into $1/8$ -pound cubes. Use the bars: how many cubes? cubes
- 3. Garden Edging.** A gardener has 12 feet of edging. Each small bed needs $1/3$ foot. How many beds can she edge?

Hint Each foot holds 3 thirds.

Show your work.

Answer beds

- 4. Frozen Yogurt.**
- A snack cart has 6 quarts of frozen yogurt. Each sample cup holds
- $1/8$
- quart. How many cups can it fill?

Hint Count eighths: 8 in each quart.

Show your work.

Answer cups

- 5. Water Bottles.** A camp has 8 gallons of water. Each bottle holds $\frac{1}{2}$ gallon. Mia says only 4 bottles can be filled. Justify or correct her answer.

Hint Ask: how many halves are in 8 wholes?

Show your work.

Answer _____ bottles

- 6. Tape Pieces.** Ivan cuts a 2-yard roll of tape into pieces that are each $\frac{1}{12}$ yard long. He gives 8 pieces to a friend.

Hint Divide first, then subtract the 8 pieces.

- a.** How many pieces does he cut? _____ pieces
- b.** How many are left? _____ pieces

Show your work.

- 7. Craft Table.** A teacher has 15 meters of yarn for bracelets.

- a.** Each bracelet uses $\frac{1}{5}$ meter. How many bracelets? _____ bracelets
- b.** Each bracelet uses $\frac{1}{3}$ meter instead. How many bracelets? _____ bracelets
- c.** Why does a larger piece for each bracelet give fewer bracelets? _____

Show your work.

- 8. Explain It.** Find $9 \div \frac{1}{3}$.

Hint How many thirds are in 1 whole? In 9 wholes?

- a.** Write the multiplication that matches this division and solve. _____
- b.** Explain why the answer is greater than 9. _____

FIND THE ERROR

explain the mistake, then fix it

- 9. Rosa's Work.** Rosa solved $4 \div \frac{1}{3}$. She flipped the first number and wrote $\frac{1}{4} \times \frac{1}{3} = \frac{1}{12}$. Explain her mistake. Then find the correct quotient.

TEST PRACTICE

choose the best answer

- 10.** Find $5 \div \frac{1}{4}$.

A $\frac{5}{4}$ **B** 20 **C** 9 **D** $\frac{1}{20}$

- 11.** Dividing 6 by $\frac{1}{2}$ gives a quotient that is –

A less than 6 **B** equal to 6 **C** greater than 6 **D** equal to $\frac{1}{2}$

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

3 X	16 V	27 E	75 D	4 Q	24 I	$\frac{3}{4}$ Z	36 D	48 I
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#3	#4	#5	#6a	#7a	#8a
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CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____